COURSE OVERVIEW

IMPLEMENT A FALL PROTECTION PLAN

The Implement a Fall Protection Plan course is designed to provide learners with the knowledge and practical skills required to correctly implement a site-specific Fall Protection Plan in accordance with legal, client, and industry requirements.

The course focuses on hazard identification, risk assessment, training requirements, equipment management, medical fitness, and rescue planning. Emphasis is placed on the responsibilities of the Fall Protection Plan Implementer, ensuring that controls are applied on site to manage fall risks and maintain a safe working environment.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

- Explain the purpose and legal requirements of a Fall Protection Plan
- Identify and control fall risks using site-specific risk assessments
- Implement training, medical fitness, and equipment management procedures
- Define and manage fall risk zones on site
- Ensure rescue plans and emergency procedures are in place
- Monitor and maintain compliance with the Fall Protection Plan

INTRODUCTION

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GRAVITY VALUES

RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.

INTEGRITY

Conducting day-to-day business with integrity at all times.

SOLUTIONS **DRIVEN**

Being solution driven.

QUALITY & **EXCELLENCE**

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

SITE ESTABLISHMENT

Before any work begins, the site must be safe.

Use this simple checklist

01 Follow the Rules:

Make sure you follow all client rules.

Example: If night work is not allowed, do not work at night.

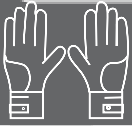
02 Rescue Plan

Have a rescue plan ready. A rescue kit and first aid kit must be on site.

03 Work in Pairs

There must be at least two workers on site.

If something goes wrong, one can help or call for help.



04 Check Equipment

All tools and gear must be safe and working before use.

05 Safe Site

Check the building or structure.

Make sure it is strong and safe to work on.

06

Manage Traffic

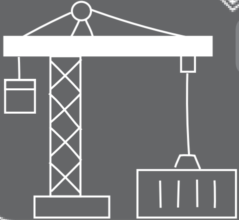
If there are cars or trucks nearby, there must be a traffic safety plan.

07

Do a Risk Assessment

Check what dangers there are.

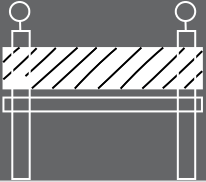
All workers must understand the risks before starting work.



08

Use the Right Safety Gear

Make sure the fall protection fits the job and the site.



EXCLUSION ZONES **& BARRICADES**

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

SITE SAFETY

- **Be healthy and ready** before starting work.
- **Only do jobs you are trained** and allowed to do.
- **Never work under the influence** of alcohol or drugs.
- **Always wear your safety gear** (PPE, harness if working at height).
- **Use harnesses correctly** and stay connected at heights.
- **Never work alone** and always follow safety steps.

SITE SAFETY

Keep people away from areas below when working above.

Don’t walk under lifted loads or stand in dangerous areas.

Only trained people can do electrical, underground, or confined space work.

Follow road rules. Wear helmets. No passengers on open trucks..

Don’t drive tired or dis-tracted.

Stop unsafe work! Everyone has the right to speak up.

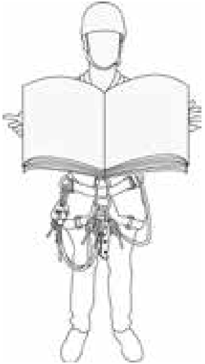

GRAVITY
GEAR

South African guidelines

In South Africa, we have the **Occupational Health and Safety Act (**OHS Act), That addresses the requirements of how to perform work safely. This includes work at height, not only Fall Arrest and Rope Access, but ANY work where a person may fall. This is referred to as a Fall Risk Position. In order to mitigate these fall risks, the OHS Act contains Construction Regulations (CR's), in particular CR 10, which require a Fall Protection Plan. The purpose of this plan is to control any and all fall risks as far as reasonably practical.

It is the purpose of this course to instruct the learner on how to implement this plan anywhere in the world where there is work done from a fall risk position.

1.1 Introduction to the Fall Protection Plan.



An FPP (Fall Protection Plan) is a document containing the admin requirements to ensure a safe work enviroment is created for all employees in a fall risk position & for the public on ground level.

An FPP is a legal requirement in South Africa, however, it has been adopted by many role players across the world and has become an Industry Best Practice. In some cases, it has become company policy.

An FPP needs to be developed by a competent Fall Protection Plan Developer, usually, that will be the Safety Officer in the organization, and it needs to be implemented on-site by the Site Supervisor/ Fall protection plan Implementor. **That is what this course is all about.**

1.2 Introduction to "Just Culture":

Just culture supports the idea of learning from unsafe acts wherever they may be observed in order to improve safety and safety awareness through the improved recognition of safety situations.

*** The "Just Culture" duties and responsibilities of all workers in a fall risk area.**
Competent persons and employees have accepted the fact that no fall risk environment can be made completely risk-free.

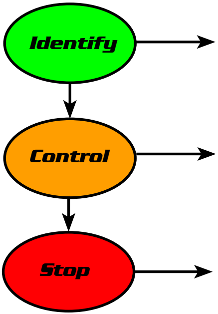
Therefore in order to implement an FPP, all employees and the company need to ensure that we have **completed the risk controls correctly.**
We all have to accept the responsibilities that the following duties require of us in order to reduce fall risks to a reasonable and practicable safe level:

1. I have a duty to avoid causing unjustified risk or harm during the task.
2. I have a duty to produce an outcome.
3. I have a duty to follow a procedural rule (SOP etc)*.

*See DSTI for the declaration and procedure that confirms the above policy.

1.3 What are the responsibilities of the Fall Protection Plan Implimenter:

- Principle 1: The Site supervisor is the FPP implementer, and needs to have the latest reversion of the fall protection plan.
- Principle 2: Admin controls (Risk assessment, Medical fitness to work at height, correct training to work at height task, equipment safe to use and fit for purpose and rescue plan if a person falls and is suspended in a harness.
- Principle 3: ID and control site-specific hazards with correct fall protection methods (see hazard register Page 6-8)



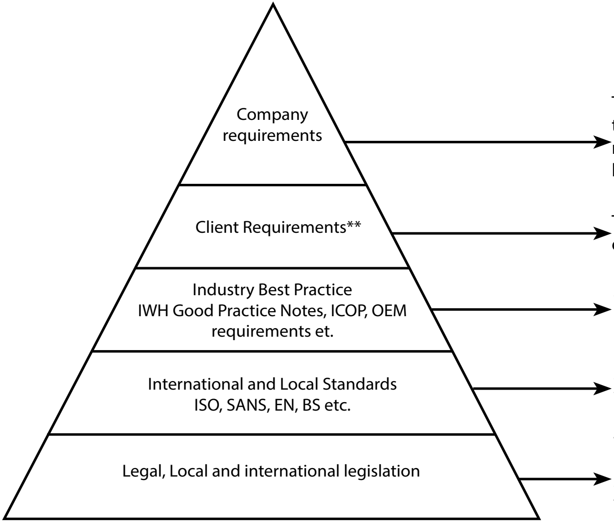
Is any person or other asset to the company exposed to potential harm?

Can the cause of the risk be avoided?
How can the risk be controlled?
Is the risk as low as reasonably practical?

If Risk can not be controlled,
Stop work and inform management.

1.4 Requirements for safely performing work in a fall risk positions.

Most countries require that every person working at height must do so in a safe manner. If your country does not have clear guidance on how to do work at height safely, you can make use of this hierarchy/ranking, by starting at the bottom of this diagram.



The employer may add any extra requirements they wish upon employees given that these requirements do not contradict any of the lower levels.

The client may have unique requirements based on the task or site which must be adhered to.

All interpretations of any legal or standard-based requirements should be aligned with what the rest of the industry agrees upon.

All equipment and techniques must comply with the relevant standards. Should no legal requirement exist, the relevant standard becomes the fundamental requirement.

All actions taken by a company must comply with the relevant legal requirements.

2. Duties of the Employer / Project manager / Contractor.

Designate a competent person to be responsible for the preparation of a fall protection plan

A competent Fall Protection Plan Developer must ensure that the entire Fall Protection Plan must be completed before any work at height may start.

This course will only be covering the actions needed AFTER the fall protection plan has been completed.

Thus it must be understood what the requirements of a Fall Protection Plan is and by extension the general requirements for performing work at height.

A fall protection plan contemplated in sub-regulation (1), must include-

1. a risk assessment of all work carried out from a fall risk position and the procedures and methods used to address all the risks identified per location;

2. the processes for the evaluation of the employees' medical fitness necessary to work at a fall risk position and the records thereof;

3. a programme for the training of employees working from a fall risk position and the records thereof;

4. the procedure addressing the inspection, testing and maintenance of all fall protection equipment; and

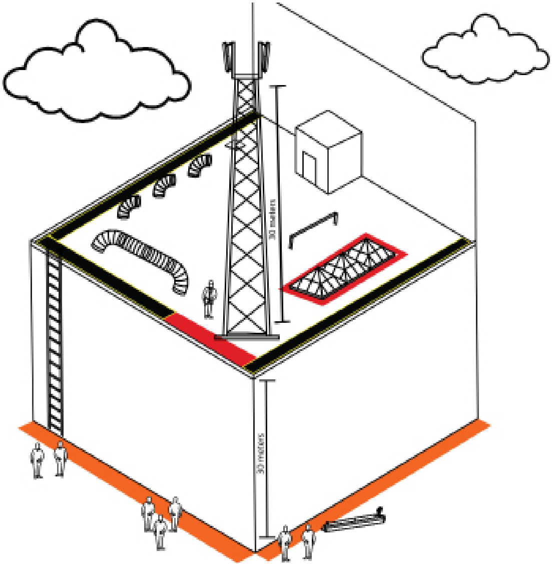
5. a rescue plan detailing the necessary procedure, personnel and suitable equipment required to affect a rescue of a person in the event of a fall incident to ensure that the rescue procedure is implemented immediately following the incident.

A contractor must ensure that a construction manager / project manager /site supervisor appointed is in possession of the most recently updated version of the fall protection plan and inducted.

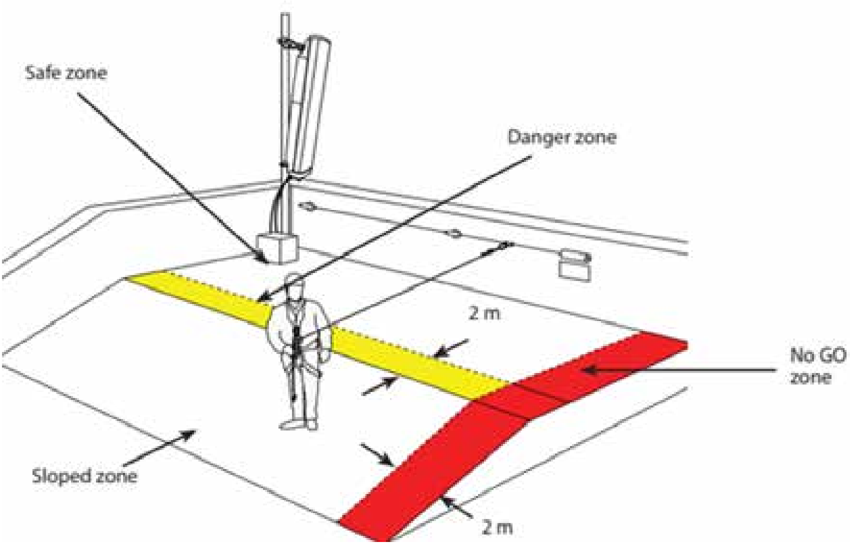
Note that the following zones will have to be taken in to consideration:

Identify, mark, and communicate the following fall risk zones before work starts:
Safe zones
Danger zones,
No go zones
Drop zones
Electricity danger zones
Point of safety and emergency assembly points

Identifying risks in any working enviroment:



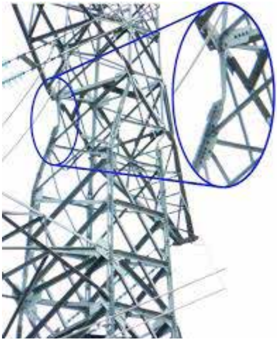
Roof Top Work Restraint System



Structural Evaluations

- "It is the responsibility of the owner of the structure to ensure that the structure is safe to climb and to perform any rigging and lifting activities when needed.
- All structures must be visually inspected before any climbing or lifting operations are conducted. It is not the responsibility of the climber or the rigger to certify the structure, but must only ensure that the structure is safe to climb. Each climber must check the following.

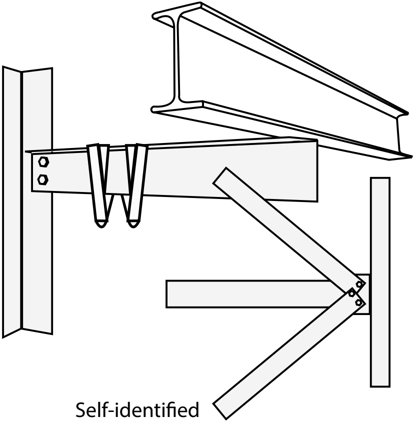
    - That the structure is singed off by an engineer.
    - That the fixed fall arrest system is certified and safe to use.
    - That there are no missing members or loose telecoms equipment.
    - Any roof structure must also be inspected that the surface areas are secure and safe to access."
    - Any planned anchor points must be pull tested annually.



Anchor Points

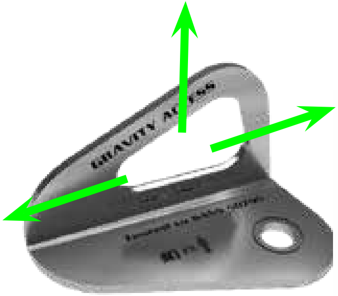
Anchor points are intended to be the last remaining connection point between a worker and a structure should a fall occur to prevent the worker from falling to the ground.
Anchor points are mainly divided into two categories:

- Self-identified anchor points, and
- Planned anchor points (EN 795:2012).

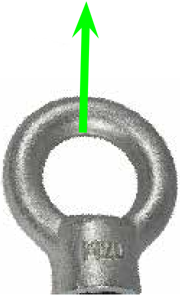


Self-identified

- S.W.L. = 150kg
- B.S. = 1500kg



Gravity Access Anchor



Eye-nut

Planned

- S.W.L. = 120kg
- B.S. = 1200kg

ALL STRUCTURES/PLATFORMS/LADDERS MUST COMPLY TO THE RELEVANT STANDARDS. THE IMPLEMENTER MUST FAMILIARISE HIM/HERSELF WITH THE REQUIRED STANDARDS BEFORE WORK STARTS.

The Fall Protection Plan Implementer will not be responsible for the certification, testing or inspection of any work platform. The Fall Protection Plan will state that the structures will be safe to work on as per the fall-risk risk assessment. The Implementer must ensure that he/she obtains proof that this is indeed the case, either, in the form of the baseline risk assessment, provided by the client, hand-over certificates, inspection registers, scaf-tags, operating certificates or engineer's permission that the site is indeed safe.

3. Planning for Work at Height

(A) Risk Assessment

Before work starts the client will have provided the contractor with a "baseline risk assessment", this will cover all risks prevalent on site, this baseline risk assessment will have been taken into account by the Fall Protection Plan Developer when writing the fall risk risk assessment, which will be the risk assessment included in the Fall Protection Plan.
It is the duty of the Fall Protection Plan Implementer to ensure that the following is identified and followed:

- **The prevalent risks on site:**
  - Falling from height.
  - Workers dropping tools or equipment.
  - Tools or equipment falling on people.
  - Dangerous areas where people are not allowed.
- **The risk controls established for the above mentioned risks:**
  - What are the needed steps to be taken in order to reduce the risks to an acceptable level. The implementer has to identify the risks and ensure that the relevant baseline risk assessment controls are implemented to ensure a safe work environment. This is indicated on the Daily Safe Task Instruction (DSTI)
- **The risk owners:**
  - The fall protection plan implementer is required to ensure that the responsibilities for implementing the risk controls are assigned as per the baseline risk assessment.
  - All risk assessment training must be recorded and the records must be kept.

Please note that, the risks on site may change daily, or even hourly depending on the site and the scope of work. The Fall Protection Plan contains a procedure for dealing with risks. It is the duty of the Fall Protection Plan Implementer to ensure that this procedure is followed. Typically, this includes the following order of risk controls:

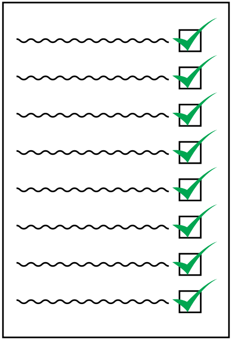
- Baseline risk assessment for site - Responsibility of the Safety officer/manager/ FPP developer
- Issue based risk assessment for task - Responsibility of the Safety officer/manager/ FPP developer
- Daily safe task instruction (DSTI) / Pre task risk assessment - Responsibility of the Site supervisor / FFP implementor

What to do when approaching a work site

RISK IDENTIFICATION

The following are factors that must be considered when identifying risks:

- **Site requirements**
- **Access method requirements**
- **Required team composition**
- **Fall protection requirements**
- **PPE requirements**
- **Environmental**



After this, the risks must be communicated to everyone on site by means of:

- Site induction
- Toolbox talk

Identifying risks on sample site (page 7)

Hazard	Assosiated Risks	Mittigation
Falling from heights	Medicaly unfit	Medical examenation of fitness that is valid, report daily medical changes.
	Incompetent worker	Work and site-specific training certificate of competence is needed
	PPE failure	Certified equipment, equipment controller, 6 monthly and pre-use inspection is needed
	No rescue plan	Rescue plan, Competent rescuer and first aider, rescue kit and first aid kit is needed
Objects falling	Tools falling	Securing tools, exclusion zones and baracation are needed
	Equipment falling	Rigging, slinging principles and baracation are needed
Open edges	Working near open edges	100% hook-up, work restraint, lifelines are needed
	Sharp edges	Edge protection, tag-lines are needed
Fragile surfaces	Falling to lower level	Barracation, work restraint, No-Go zones and good housekeeping are needed
Dangerous Environmental conditions	Adverse weather, Insects & animals, smoke	Stop work procudure in adverse weather conditions, Emergency preparness plan for insects and animals, propper PPE, Fire extinguisher.
Radio Freqency radiation	Over exposure to RF radiation.	Power down procudure, only RF trained workers, make use of RF scanner, do not work in red zones.

Hazard Register

HAZARD EXPOSED TO WHILE WORKING	CONSEQUENCE (worst case scenario)
Open edges	Falling over the edge to a lower level
Holes, openings and excavations	Falling into, and causing serious injury and or death
Fragile surfaces	Falling through to a lower level or into plant or moving machinery
Steep slopes or ramps	Sliding down slopes generating enough speed to cause injury or fall over the open edge at the end of slope
Potentially unstable surfaces or structures	Danger of engulfment or structure failure
Level changes	Falling to a lower level
Cluttered work area	Slip and trip over edge or into a hole or causing dropping objects
Lack of proper hand grip	Losing balance and falling AND or difficult access and egress from large plan or vehicles causing a slip and fall to a lower level..
Sharp edges or hot pipes and plant	Cutting or severing PPE during a fall arrest event

Loose objects or tools that may fall	Falling to a lower level causing injury or damaging plant
Moving plant or machinery	Causing you to lose your balance or hitting you or your PPE while you are suspended after falling
Unsafe entry and exit to work area	Cannot enter or exit work area safely and or cannot move around safely in work area with exposing self to falling or falling objects.
FALL PROTECTION HAZARD	CONSEQUENCES
High Shock load	Sudden stop after a fall, causing serious injury or death e.g. Using static or semi static lanyard as fall arrest PPE this will not absorb the energy generated by your fall
High fall factor	Will result in long free fall distances
Sliding anchors	Will cause connectors to break and result in high shock loads
Not enough free space for fall arrest	Will result in person hitting obstructions before the fall arrest system can activate
Swing back	Hitting the structure support
Swing down	Fall onto a lower level or the ground

Overreaching from behind passive fall protection	Centre of gravity is outside the safe work area and may cause a fall over the passive protection resulting in a fall to a lower level causing serious injury, death, or damage to property
Working near electricity lines	Access method or person touching the live electricity may result in serious injury, death and or damage to property
Not using of access method as per OEM safe use criteria	Resulting in access method failure or person falling from access method causing serious injury, death or damage to property

Example of a Baseline Risk Assessment

DATE:	FALL RISK ASSESSOR NAME:	NAME OF TASK SELECTED:	WHO AND OR WHAT IS AT RISK?:

ADMINISTRATIVE HAZARD (CR10(2) (b, c, d, e) LIST A MINIMUM OF AT LEAST 4 ADMINISTRATIVE HAZARDS							
Hazard: *The source or situation with a potential to cause human injury or ill-health or harm*	**Possible Harm and or property damage**	Likeli hood	Severity	Risk	**Controls and Preventative measures** Who? When? What or How?	Likeli hood	Severity	Risk
1. **Working while medically unfit:**	Medical conditions may cause a fall from a height or endanger others while rescuing the person, Serious injury, or death.	3/4/5	5	15/ 20/ 25 H	**Sec16(2)** Appoint FPP D who must develop a procedure for the evaluation of medical fitness Site Supervisor Ensure that the process for medical evaluation has been completed – Check COF are valid before work starts. Manage DSTI process and allow an opportunity to report medical unwellness by an employee before work start. (Employee has a duty to inform)	1	5	5 **Med**

The DSTI has to be completed every day before work starts, by the Site Supervisor, this may be the same person as the fall protection plan implementer. If it is not however the DSTI must be checked by the implementer to ensure that it has been completed correctly. An example of how a DSTI must be completed is given on the next page.

Example of a DSTI

Site Area		Date	
Site Task Area			
Permit number(s)			
Work at Height Risk Assessment No			
Supervisor Name		Contact No	
SCOPE OF WORKS			
Describe task to be performed:			
List the relevant steps of the task			
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	
First Aider/Rescuer:	1:	2:	3:
EMS No:		Police No:	
Fire/Rescue Team No:		Company Contact Details	
How will EMS access site:			
SITE SPECIFIC RESCUE PLAN			
Rescue method	☐ First Aid Kit	☐ Rope Grabs	☐ Descending Device
	☐ Ratchet Rescue Kit	☐ Rescue Slings	☐ RA lift and lower kit
	☐ Stretcher and basket	☐ Plug-in for tensioned lines	☐ RA lowering kit
	☐ Cableway rescue kit		☐ Rig for rescue
Note CRB(7) and/or competent rescuer on site to discuss the rescue procedure on site as per the task requiremets			

All of this is accomplished with induction training and Daily Safe Task Instructions or DSTI's.


GRAVITY

(B) A process for determining medical fitness of workers.

Part of the risk assessment has to address the risk of falling from height, one of the causes of this could be medical unfitness thus there must be a process for determining whether a person is medically fit to perform the specified work. As well as there must be proof of said medical fitness. This usually comes in the form of a Certificate of Fitness (COF) in the form of Annexure 3 of the Construction Regulations. The Certificate of Fitness must contain the following details:

- The employees full name and surname.
- The employees ID number.
- The date of the examination.
- The date of expiry, this may not exceed 1 year from the date of examination.
- The result of the examination, fit or unfit as per job specification.
- Must be signed off by an Occupational Health Practitioner.

In the same way that the risk assessment may be subject to near constant change, so can the medical fitness of workers be subject to rapid and unpredictable change, based on the site and the scope of work.

It might be required that workers are subject to more frequent medical evaluations than the minimum annual examinations. This may be due to confined space work, toxic gasses or hazardous chemicals.

The Fall Protection Plan will include the correct procedure relevant to the site and scope of work. It is the duty of the Fall Protection Plan Implementer to ensure that this procedure is followed by everyone on site and that the COF's are available on request, as well as any other required proof of this implementation.

ANNEXURE 3

OCCUPATIONAL HEALTH AND SAFETY ACT, 85 OF 1993 CONSTRUCTION REGULATIONS, 2014

MEDICAL CERTIFICATE OF FITNESS

Name of Employee ______________________ ID Number ______________________ Co Number __________

*Occupation e.g. General Worker, Welder, Bricklayer, Steel fixer, Mobile Crane Operator, etc.	*Possible Exposures e.g. noise, heat, fall risk, confined space, etc.	*Job Specific Requirements e.g. Operating Mobile Crane, Digging Trenches, Erecting Form work & Support work, etc.	*Protective Equipment e.g. Dust Respirator (light duty), Welding Gloves, etc.

***The employer to complete the information in the spaces marked with an * before sending the Employee for a medical examination**

Declaration by the Medical Examiner

I certify that I have, by examination and testing, using the above criteria specified by the employer, satisfied myself that the above mentioned employee is fit to perform the duties as described by the employer in the matrix above.

Occupational Medicine Practitioner / Occupational Health Nursing Practitioner: (Please Print Name) ______________________

Signature ______________________ Practice Number ______________________ Date ______________________

Address ____

(C) A Training programme.

To ensure that all workers on site are aware of the correct method to use in order to perform a given task safely, workers must undergo training.

The required training is not limited to Fall Arrest or Rope Access training, but includes any and all training required for a worker to perform his/her task effectively, competently and, above all, safely.

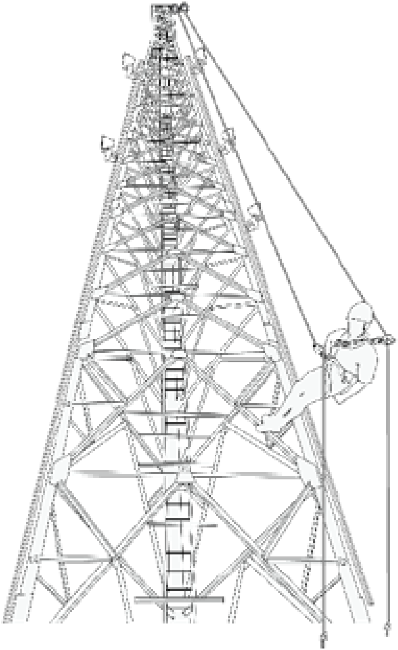
The Fall Protection Plan will include a "training programme" which will stipulate the training required as per job specification. As well as when and how often the training must be received.

Typical Work at Height training includes, but is not limited to:

- Basic Fall Arrest .
- Fall Arrest Basic Rescue.
- Advanced Fall Arrest Rescue.
- Rope Access Technician.
- Rope Access Practitioner.
- Rope Access Supervisor.
- Rope Rigging.
- Mechanical lifting.
- Mobile elevated platforms.
- Temporary suspended platforms.
- Scaffolding.
- Climbing Equipment inspection management.
- Lifting tackle inspector.
- Ladder use
- Scaffolding errectors / inspectors

- Telecommunication Tower Abseiling

- Rope Rigging is a course that teaches one how to properly sling loads / equipment and use the correct systems to lift the load in a safe manner.



Telecommunication Tower Abseiling

The nature of telecommunication towers do not always allow for fall arrest equipment to be used in a safe or effective manner. Therefore, it might be necessary to descend from the top of the tower using rope access to reach the required work area. The following must be adhered to when using rope access on site:

- At least two workers with telecommunication tower abseiling certificates on site
- Be on two points at all times
- No ascending equipment to be used
- Always rig for rescue

Apart from access method specific training, all workers must also receive training on the Fall Protection Plan, or what is referred to as FPP Induction, this includes:

- Risk Assessment.
- Emergency Procedures.
- Safe Operating Procedures.

The Fall Protection Plan will include the frequency at which these trainings must be conducted, some might be every 3 years, some might be required more than once a day. It is the duty of the Implementer to ensure that there is proof that the required training has been conducted at the required intervals as set forth by the fall protection plan.

Proof of all training must be available on request, this may be given in the form of certificates, induction registers, or attendance registers
On the next page is an example of a "training matrix" which specifies the typical yearly training requirements as per typical designations.

Example of a training matrix

No.	Designation	Training Requirements
1.	FPP Developer	UUS 98&95 (Fall Arrest and basic rescue) 229994 (FPP Developer)
2.	FPP Implementer	US 98&95 (Fall Arrest and basic rescue), FPP Implementation
3.	Site Supervisor	US 98&95 (Fall Arrest and basic rescue) FPP Induction, First Aid
4.	Rope Access Technician - Level 1	US 98&2300009 (Rope Access lvl 1), FPP Induction
4.	Rope Access Practitioner - Level 2	US 96(Rope Access lvl 2), FPP Induction, First Aid
5.	Rope Access Supervisor - Level 3	US 97&230001(Rope Access lvl 3), FPP Induction, First Aid
6.	Equipment Controller	Climbing equipment inspection and management, US 98&95 (Fall Arrest and basic rescue) FPP Induction Relevant OEM training
7.	First Aider & Rescuer	US 98&95 (Fall Arrest and basic rescue), FPP Induction, First Aid
8.	Basic Fall Arrest Technician	US 98(Basic Fall Arrest), FPP Induction
9.	Fall Arrest and Basic Rescue Technician	US 98&95 (Fall Arrest and basic rescue), FPP Induction, First Aid
9.	Rope Rigger	US 98&95 (Fall Arrest and basic rescue), 14706, FPP Induction
10.	Advanced Fall Arrest Rescue Coordinator	US 999(Advanced Fall Arrest Rescue) , FPP induction, First Aid
11.	Visitor	FPP Induction*

(D) Equipment Management

Referring back to the risk assessment, another cause of people, tools or equipment falling might be equipment failure. To prevent this, there must be a procedure in place for the inspection, testing and maintenance of equipment.

All equipment used on site must be inspected in intervals specified by the manufacturer or the relevant standards. ALWAYS FOLLOW OEM REQUIREMENTS FOR EQUIPMENT.

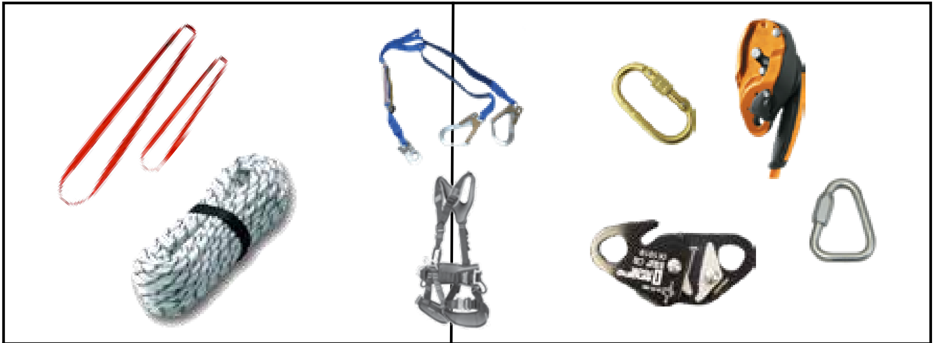
According to ISO 22846 work at height equipment must be inspected every 6 months, before, after and during use, as well as daily and interim inspections are needed. However it is considered industry best practice to inspect equipment every 3 months. The relevant inspection criteria, storage requirements and replacement procedures of equipment is given here for the benefit of the fall protection plan implementer in order to provide a basis for the implementer to be able to make a judgement on the sufficiency of the equipment management.

All requirements, policies, procedures, and responsibilities for equipment management must be documented and recorded in the Fall Protection Plan and it is the duty of the Fall Protection Plan Implementer to ensure they are implemented., and that gear gets inspected before use.

It is important to note that although the Construction Regulations call for an equipment testing procedure, no PPE may be subject to load test, drop test, or any form of destructive testing and then subsequently be used again

Should companies wish to test equipment, refer all requirements to the OEM and record all relevant procedures in the Fall Protection Plan.

Inspection Criteria:



Software	Hardware
• Condition of stitching • Discolouration • Wear and tear • Cuts • Burns • Date of manufacture • (for rope) Condition of core	• Rust • Deformation • Working action • Movement • Spring action • Wear and tear

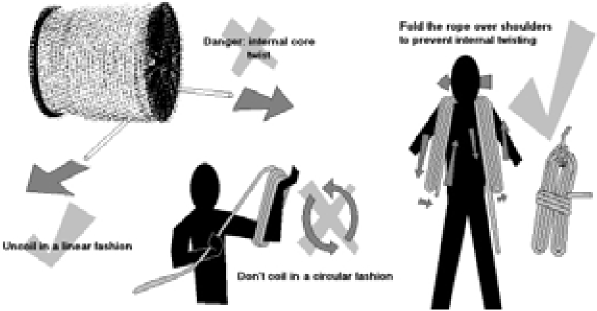
Example of Pre- use inspection form

PPE-USE INSPECTION SHEET					
No:	Equipment	Safe to Use? Y/N	User (Responsible Person)	Signature	Date
1					
2					
3					

Store Equipment:

- In a cool dry place - This will help prevent mildew in software and rust in hardware.
- Free of chemicals - This will reduce the chance of chemical damage to equipment.
- Where there are no infestations (rats and moths etc.), which can damage software; ensure that there is a system in place to prevent this.
- In a secure location - This will prevent unauthorized personnel entering and potentially damaging equipment.

Rope Care

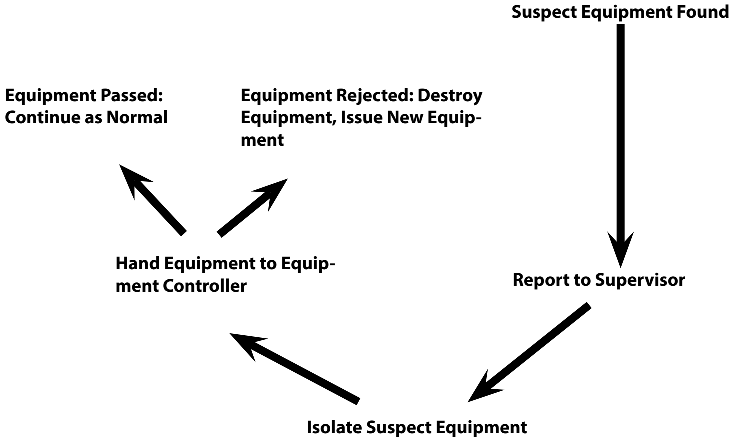


- Do not use rope for anything other than work at height.
- Do not run rope over sharp edges, use edge protection.
- Do not stand on or walk over a rope.
- Ensure that all devices are compatible with any devices to be used.

Reporting Suspect Equipment

Equipment that is found to be damaged or too old for use must be reported to the person responsible for the site (either the Site Supervisor or Safety Officer etc).

Damaged or expired equipment must be properly marked and removed from service immediately. Damaged equipment must be destroyed at the earliest opportunity to prevent the equipment getting back into the work system.



Repairing Damaged Equipment

Equipment may only be repaired by either the manufacturer or a company certified by the manufacturer. **NO CONTRACTOR MAY REPAIR PPE.**

All the procedures pertaining to equipment will be included in the Fall Protection Plan, it is the duty of the Fall Protection Plan Implementer to ensure that these are being followed on site. This will include requesting inspection records, rejection notices, requisition forms etc. to ensure that the workers on site are complying to the relevant procedures.

(E) Rescue Plan

Before any work may start on site, one must consider the worst case scenarios and this includes all emergency procedures as described in the HSE file, as well as the site-specific fall rescue plan.
It is the duty of the FPP Implementer to ensure that:

- All workers on site are aware of the rescue to be conducted
- The rescue plan is applicable to the site
- There must be a competent rescuer and a first aider on site

The Fall Protection Plan will include a Rescue Plan, this plan will be site and task specific and will be written to the competencies of the rescuers on site. It is the duty of the Fall Protection Plan Implementer to ensure that the necessary equipment and personnel are present on site should it be needed to perform a rescue and that all workers are aware of the relevant emergency procedures. Proof that the workers are aware of these procedures and that all the necessary equipment and personnel are available on site, will be given in the form of induction training registers, DSTI's, and rescue training certificates.

The rescue plan must be written by a competent Subject Matter Expert (SME).

Example of an emergency procedures:

Rescues

Ensure that a rescue plan is in place to ensure that a rescue can be performed as quickly and efficiently as possible.

Perform a rescue when;

- A person has fallen and his shock-absorbing lanyard has deployed.
- A person has lost consciousness.
- A person is unable to climb down to safety unassisted.

All rescues are different but there are a few things that stay the same throughout:

- Always call the emergency services before attempting a rescue.
- Always establish and maintain communication with the casualty throughout the rescue with either: two-way radios, hand signals or simply talking if the person is within earshot.
- It is recommended that all rescuers are trained first aiders as well.

Ratchet rescue procedure:

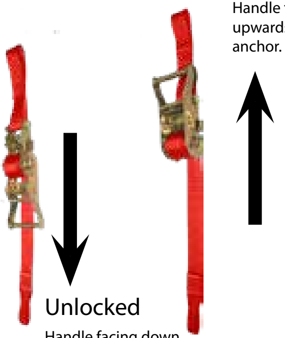
1. Call the emergency medical services.
2. Climb up to the casualty.
3. Connect the rope to the casualty's top D-link.
4. Connect the suspension trauma sling.
5. Climb above the casualty.
6. Anchor the rescue ratchet with an attachment sling.
7. Remove all the slack between the casualty and the descending device.
8. Lift the casualty with the rescue ratchet.
9. Remove the casualty's lanyard.
10. Apply extra friction.
11. Lower the casualty.

Example of a Rathcet rescue kit

1. Attachment sling
2. Rescue ratchet
3. Descending device
4. Kernmantle rope
5. Suspension trauma sling
6. Attachment karabiner
7. Extra-friction karabiner
8. Kit bag

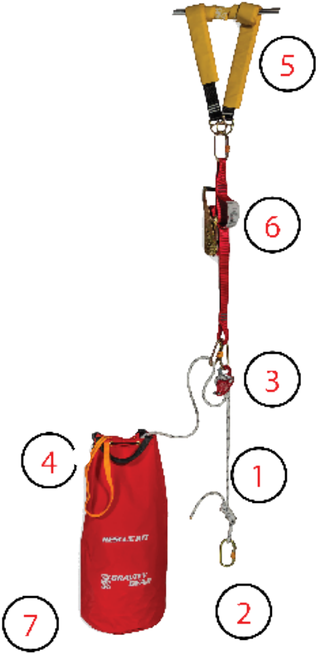
Locked

Handle facing upwards towards the anchor.



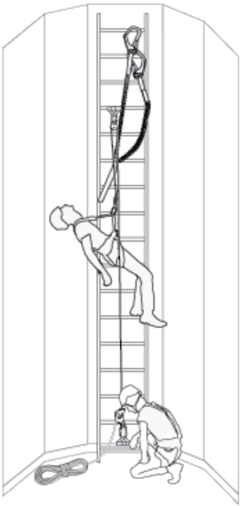
Unlocked

Handle facing down towards the load.

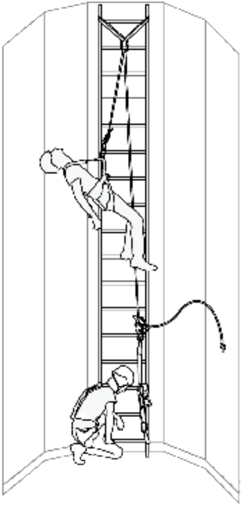


Example of a Monopole rescues

Rescue kit installed up-side down

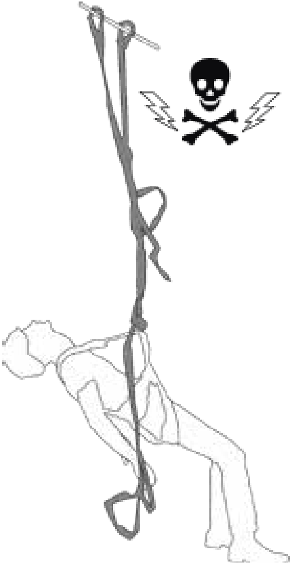


Rathcet rescue kit installed up-side down



Associated risks while awaiting a rescue

Suspension Trauma



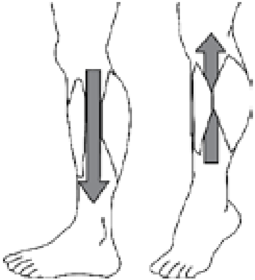
The two biggest risks associated with awaiting a rescue while suspended in a harness, are injury or death due to suspension trauma.

Due to the force of gravity, the blood in the fallen person's body is pulled toward the ground, and pools in the legs.

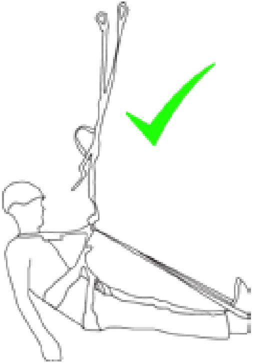
Due to the constriction of the leg loops of the harness, the blood cannot return to the fallen person's heart. This prevents blood being circulated through a person's body. Thus, less and less blood reaches the brain, leading to loss of consciousness and can lead to brain damage. Finally, this can lead to cardiac arrest and eventually death.

All of this can happen within 15-20 minutes; so it is important to have a good knowledge of the techniques needed to prevent suspension trauma, for yourself, and the casualty that needs to be rescued.

Preventing Suspension Trauma



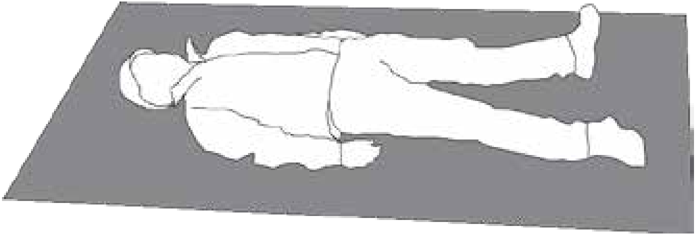
Muscle Pump: Moving your legs, feet and toes will contract the muscles in the legs and force blood back into circulation, which provides the brain with the much needed oxygen to stay conscious.



Suspension Trauma Sling: Will allow you to stand up or to get horizontal, prevent the blood pooling in the legs as well as relieve the pressure of the leg loops which will allow the blood to remain in circulation to the heart and brain. Should the casualty be unconscious, the rescuer can do this for the casualty.

Recovery Position

This position allows the blood to flow slowly enough as to prevent Re-flow Syndrome, while at the same time keeping the casualty stable and still as to prevent any possible further injury to the casualty's neck and/or spine. This position is also known as the "Supine Position".



(F) Review Plan

Ensure that the fall protection plan contemplated in paragraph (a) is implemented, amended where and when necessary and maintained as required; and
take steps to ensure continued adherence to the fall protection plan.

A Fall Protection Plan must be reviewed to ensure that it is still up to date and that it still manages all fall risks on site.

The FPP Implementer need not review the plan him/herself, but it is however the duty of the implementer to monitor the plan and inform the FPP Developer. Should the plan become insufficient or outdated, or if one of the following should occur:

- Should there be a change in legislation.
- Should there be a change in work scope.
- After an incident has occurred.
- After one year has passed.
- Should the client request it.

(G) Incident Reporting.

Any and all incidents must be reported.
Using the relevant document templates provided.

Should there be any operation that falls with outside the relevant accepted standards (e.g. operations outside the OEM requirements) work must be stopped and the transgression reported to the Construction Manager/Project Manager/Risk Assessor/Safety Officer/HSE Representative.

Any and all reported transgressions must be documented and recorded with as much detail as possible. This will aid the investigation, should there be one, as well as allow the cause of such transgressions to be identified and removed.

Should the Fall Protection Implementer be unsure of the course of action he/she must contact the fall protection plan developer in order to determine the best way forward.

THE WAY FORWARD

Congratulations on completing Gravity Training's Implement a Fall Protection Plan course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map or consult your facilitator/assessor.